Moving Aircraft Detection in High-Resolution SAR Using Velocity-Induced Phase Information

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***Abstract*—Despite recent advances in deep learning-based aircraft detection, moving aircraft detection in synthetic aperture radar (SAR) remains challenging due to motion-induced phase distortion. The number of moving aircraft in a satellite SAR image is significantly smaller compared to that of nonmoving aircraft, complicating the acquisition of training data. Accordingly, this study proposes a robust moving aircraft detection framework for high-resolution SAR images, leveraging: 1) augmented training data exploiting the Doppler effect of aircraft motion and 2) a 2-D module designed to effectively capture motion-induced phase distortion. First, Doppler effects caused by the azimuth and range velocity of aircraft are considered and applied for training data augmentation. In addition, motion-driven features of aircraft are enhanced by the proposed 2-D module in the object detection algorithm. Evaluation of the proposed aircraft detection is performed synergistically using Umbra SAR images and real-time automatic dependent surveillance broadcast (ADS-B) data, collecting 11 275 aircraft from SAR images. The detection results on moving aircraft are assessed using $AP_{0.5}$ and scored 0.855, demonstrating detection enhancement of 5.5% compared with conventional object detectors. For nonmoving aircraft on the other hand, $AP_{0.5}$ scored 0.934, which is regarded as more accurate than the majority of aircraft detection algorithms using high-resolution SAR images. The proposed detection algorithm and augmentation strategy are able to be applied to aircraft motion estimation by measuring its location and velocity.**

***Index Terms*—Aircraft target detection, automatic dependent surveillance broadcast (ADS-B), ground moving target indication (GMTI), moving target, synthetic aperture radar (SAR).**

I. Introduction

MONITORING aircraft is critical for target reconnaissance and monitoring, as its outcome derives critical information for decision-making for military purposes. As an active sensor, synthetic aperture radar (SAR) acquires images regardless of weather conditions and sunlight by transmitting high-frequency microwave [2]. Target detection using SAR image has been widely studied recently; traditionally, threshold-based detection algorithms such as the constant false alarm rate (CFAR) method have been applied to target identification by detecting anomalies against background clutter [2], [3]. As targets identified from CFAR based on differences with neighbor cells, their accuracy is relatively low under a complex background clutter environment. In addition, an aircraft detector directly exploiting its geometric feature was proposed by combining a canny edge operator and clustering [4]. While traditional detection algorithms demonstrated high performance under controlled experiments, their practical application is limited due to its dependance of prior information from target characteristics [5].

As high-resolution SAR images become increasingly accessible, object detectors based on deep learning are widely applied, demonstrating accurate performances compared to conventional algorithms [6], [7], [8]. The convolutional neural networks (CNNs) are regarded as competent in extracting specific image patterns, with the availability of open-source SAR datasets [7], [9], [10]. To refine de-noised textual features, Wang et al. [11] fused image features and frequency features using a feature fusion block, increasing detection accuracy, particularly in inshore regions. Tian et al. [12] integrated three different convolution networks into You Only Look Once (YOLO) v5 to improve small target detection. It accordingly reduced processing time while simultaneously enhancing the detection accuracy. In addition, recent studies employing CNN-based detectors utilize attention modules to capture contextual information and emphasize salient regions of the target [13], [14]. Since the attention mechanisms could enhance feature representations by assigning greater weight to the most relevant regions in the feature maps using similarity, they are widely adopted in small target detection to highlight target-specified characteristics [15], [16], [17].

Additionally, after the advent of the transformer architecture in natural language processing (NLP), the vision transformer and detection transformer (DETR) were developed, incorporating self-attention mechanisms to enable global

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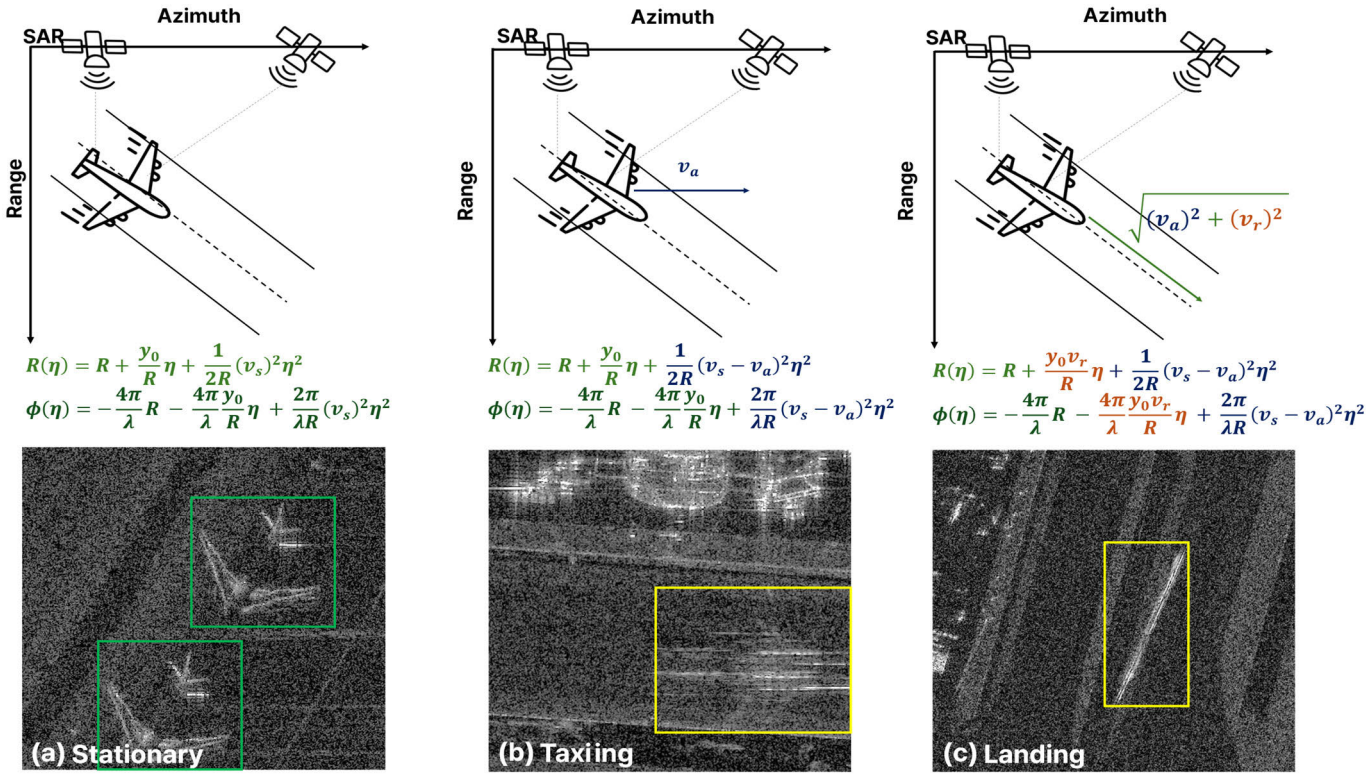


Fig. 1. Illustration of SAR acquisition geometry and Doppler phase equation with examples of (a) a stationary aircraft, (b) a taxiing aircraft with azimuth velocity, and (c) a landing aircraft that has azimuth and range velocity both.

interpretation of targets [18], [19]. These detectors demonstrate high performance in capturing global contextual information compared to conventional CNN-based detectors. The self-attention mechanism enhances image context understanding by computing attention weights through correlations between query and key components, which are accordingly applied to value components derived from the feature map.

Transformer-based detectors for SAR targets are widely introduced for additional enhancement of target detection performances [20]. To achieve high accuracy in target detection using an oriented bounding box, Feng et al. [21] developed an orientation enhancement module. As small vessels are often indistinguishable due to their scale and background noise, Qin et al. [22] proposed an algorithm that refined queries corresponding to small ships using complete contrastive denoising (CCD) and extracted accurate target features through look twice toward maximum (LTTM).

Despite such enhancements, conventional deep learning algorithms have predominantly focused on detecting nonmoving targets. The signatures of moving targets in SAR images are distorted or relocated due to additional velocity induced by phase terms [23]. In the case of aircraft, their range of motion is significantly varied by their state of takeoff, landing, taxiing, and nonmoving. As in Fig. 1, the signatures of aircraft are significantly different depending on their velocity and flight conditions. These variations complicate feature extraction and the detection of aircraft in SAR images. Furthermore, a limited number of aircraft data are able to be acquired with motion, leading to a training data imbalance in terms of quantity. Although deep learning-based detectors require sufficient training data for accurate detection performance [24], little research was conducted on developing datasets specifically designed for moving aircraft detection. Consequently, these limitations result in inaccurate or missed detections, which in turn degrade the overall performance of moving target detection and identification.

The automatic-dependent surveillance broadcast (ADS-B) system is widely used for tracking the aircraft flight state both within land and aerial space [25]. However, aircraft without an ADS-B signal, especially when engaged in military operations, cannot able to be tracked using ADS-B data. Additionally, when the aircraft moves within the airport during taxiing or take-off preparation, apron and other man-made target signals interfere with the ADS-B signal acquisition.

Accordingly, this study proposes an advanced transformer-based detector using an augmented training dataset generated from target motion-induced phase information. To refine the features of moving aircraft, we design a 2-D (Deformable + Difference) module that captures the scattering characteristics of the target's motion-driven phase. The 2-D module highlights low-level features by adapting to diverse target representations and extracting edge information distorted by aircraft motion. Moreover, phase-distorted targets are generated to remedy the data imbalance and quantitatively augment the training data using an SAR phase refocusing function [26] and a diffusion-based generative model [27]. This approach synthesizes simulated target phase distortions induced by azimuth and range velocities. To the best of our knowledge, this is the

first attempt to explore the recognition of aircraft motion state using real SAR imagery.

This article is organized as follows. Section II represents a review of current research work in the field of SAR aircraft detection and moving target indication algorithms. Section III introduces the acquisition process of our experimental data and proposed algorithms in detail. The experiment on the proposed method is validated in Section IV. Finally, the conclusions are presented in Section V.

II. Related Works

A. Deep Learning-Based Methods for SAR Aircraft Detection

SAR-based aircraft detection and recognition have been extensively studied, evolving from traditional feature-based approaches such as CFAR detectors and geometric scattering models to modern deep learning frameworks [28]. Early methods relied on handcrafted features derived from the physical scattering characteristics of aircraft structures, but were limited in generalization under complex clutter environments. With the advent of deep learning, both detection accuracy and recognition capability have improved significantly, motivating a broad range of architectural innovations. However, aircraft detection in SAR images remains particularly challenging compared to vessel detection, due to the presence of complex clutter such as boarding bridges and aprons [29]. These structures introduce different backscattering patterns and result in distinct signatures as influenced by the geometric configuration of the SAR acquisition. To address these challenges, many researchers have conducted studies to improve detection accuracy by means of advancements in network architecture. Li et al. [29] designed an informative correction block (ICB) using a salient point to suppress background amplitudes. This proposed module demonstrated effectiveness in mitigating interference from clutter. Taking the mechanisms of multilook, Zhang et al. [30] developed coherent scattering enhancement (CSE) for extracting fine aircraft target features in deep neural networks. By increasing the strong coherence of artificial targets, this strategy enhanced the signal-to-clutter noise ratio (SCNR). Considering that complicated backscatters induce high speckle noises, a despeckle enhancement transformer network (DET-net) was proposed and demonstrated advances in peak signal-to-noise ratio (PSNR). To address the dispersion of point features by neighboring pixels, Chen et al. [31] developed an incentive attention mechanism using incentive channel attention. In addition, the researchers used geometric characteristics of aircraft targets by designing additional networks. Kang et al. [32] introduced SFR-NET by exploiting the geometry of aircraft components, such as tails, wings, and engines, based on contextual feature attention. Zhao et al. [8] developed an attention feature fusion module incorporating split attention mechanisms and deformable lateral connection (DLCM) for capturing abundant aircraft features. In [33], scattering information enhancement was conducted with a hybrid approach that incorporated edge information and Gaussian mixture model modeling. Wang et al. [34] refined aircraft's morphological features with a morphological feature-aware module, combining spatial attention and channel-pooling. Meng et al. [5] improved localization performance by integrating scattering key points and relative coordinates to refine structural representation. To leverage scattering features, Huang et al. [35] designed a scattering information extraction and enhancement module (SIEEM) incorporating scattering extraction and scattering enhancement. Luo et al. [36] integrated spatial-frequency domain joint learning via fractional Gabor transform to capture orientation-aware features and enhance aircraft saliency across varying aspect angles.

Deep learning-based detectors require large volumes of accurately labeled data to achieve robust performance. For dealing with this problem, Zhang et al. [10] adopted multiscale feature fusion for robust detection of targets with diverse scales and established a benchmark dataset. Further, to solve the class imbalance problem, Bao et al. [37] used class-balanced data augmentation (CC-DA) instead of resampling-based augmentation and represented advanced performance in small-sampled classes. Zhao et al. [38] introduced global instance contrast to maximize small interclass similarities by adopting contrastive loss. Unlike other detectors, Zhou et al. [39] first developed a diffusion-based detector, demonstrating the effectiveness of generating a mechanism from the diffusion process. For lightweight aircraft detection, Wang et al. [40] designed MSADNet with a reduced number of parameters, incorporating a multiorder attention module to improve the backpropagation while minimizing computational overhead. In addition, several studies have also utilized aircraft geospatial context to enhance detection performance. Xiao et al. [41] reduced high false alarms by extracting potential areas and achieved efficiency in inference. Luo et al. [42] developed a fine-grained detection framework by leveraging prior geospatial information and a geographic category distribution map. Considering the characteristics of azimuth angles from SAR, Meng et al. [43] fused subaperture and full-aperture SAR by a Siamese backbone to refine angular-specific scattering features. Despite the current attempts in aircraft detection, they tend to overlook shape distortions caused by target motion. Although a few methods considered the defocusing effect of the target, they primarily focused on low-velocity scenarios, without accounting for severe defocusing caused by high-speed motion or azimuth offsets induced by range-direction velocities. Above all, the acquisition of accurate training data under different background or motion conditions is not considered, nor is it validated with reliable ground truth.

B. Moving Target Detection in SAR

To find moving targets in SAR, ground moving target indication (GMTI) has been widely applied. The GMTI techniques basically recognize moving targets using the above two-channel SAR data. Displaced phase center antenna (DPCA) estimates regions from motion state targets, highlighting the moving target from background clutter by subtracting two SAR image amplitudes. Cerutti-Maori and Sikaneta [44] proposed an extended DPCA that incorporated the clutter-plus-noise covariance matrix for clutter suppression, demonstrating accurate detection of moving targets. Huang et al. [45] improved GMTI systems using joint-pixel DPCA, refocused with high-order term effects, and relocated moving targets by

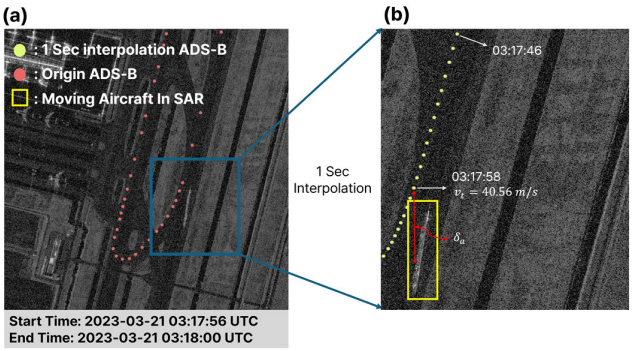


Fig. 2. Example of generating moving aircraft validation data by matching interpolated ADS-B data. (a) Moving aircraft with origin ADS-B data and (b) 1 s interpolated ADS-B with calculating target velocity (v_t) and azimuth offset (δ_a).

TABLE I
SPECIFICATION OF UMBRA SATELLITE USED IN THIS STUDY

Radar Sensor Parameter	Value
Center Frequency	X-band (9.65×10^9Hz)
Chirp Bandwidth	1.2×10^6Hz
Polarization	HH or VV (Single Pol.)
Resolution	0.25m – 1m
Swath width	5km × 5km

error-weighted subspace projection (EWSP) in multichannel SAR. Thanks to the applicability of deep neural networks, deep-learning-based GMTI methods were examined. Mu et al. [46] developed CV-GMTINet for the GMTI task using a complex-value CNN framework, which enhanced moving target features through a multichannel SAR fusion process. In addition, recognition of moving targets with shadows in multiframe SAR was studied. Ding et al. [47] utilized target shadow information for moving target detection in the frames of video SAR and combined Bi-LSTM and Faster-RCNN to reduce missed detections. Bao et al. [48] developed ShadowDeNet, which refined features from moving target shadows regardless of deformable sizes and background clutter. Moreover, Yan et al. [49] detected moving targets by emphasizing morphological characteristics such as azimuth-defocused patterns, using refined feature maps.

Conventional GMTI approaches have contributed to the detection of moving targets in high-resolution SAR imagery. However, the methods such as DPCA typically require multichannel SAR systems, limiting their applicability to current satellite platforms. Similarly, deep learning-based techniques often rely on multichannel configurations or are only validated in controlled environments such as simulations, thereby restricting their practical deployment. Given the physical properties associated with moving aircraft, incorporating a phase-sensitive augmenting strategy along with a neural network designed to emphasize such characteristics may be one of the effective approaches for detecting across a wide range of target velocities.

III. DATA AND METHODOLOGY

A. Data Acquisition

The dataset used in this study comprises aircraft targets acquired in the Umbra SAR satellite image, including single look complex (SLC) and geo ellipsoid correction (GEC) products. The SAR data were acquired in X-band spotlight mode, and the detailed specifications of the Umbra SAR imagery are presented in Table I. The labels of stationary aircraft were acquired through an expertized visual inspection based on prior knowledge and electro-optical satellite image comparison. On the other hand, the moving aircraft dataset was validated using the ADS-B acquired from FlightRadar24 [50]. In the ADS-B data, time, longitude, latitude, altitude, speed, and direction were recorded for each aircraft [represented in Fig. 2(a)]. To enable accurate data fusion with SAR imagery, the ADS-B data were temporally interpolated at one-second intervals as illustrated in Fig. 2(b). Subsequently, based on auxiliary metadata from the corresponding SAR images, we extracted ADS-B records within a 20-s window centered on the SAR acquisition time, which is from 10 s before to 10 s after the center time, to ensure precise temporal alignment. Through this process, both the position and velocity of the target were accurately synchronized with the SAR acquisition time. Additionally, azimuth offsets caused by range direction velocity components were incorporated into the temporally aligned ADS-B data. When spatial overlap was observed between moving aircraft in the SAR and the synchronized ADS-B tracks, the matching and validation were performed. The total process is illustrated in Fig. 2.

TABLE II
DIVISION OF THE SELF-BUILT UMBRA SAR DATASET

Category	Train	Val	Test
Stationary Aircraft	7211	1916	1245
Moving Aircraft	633	165	105

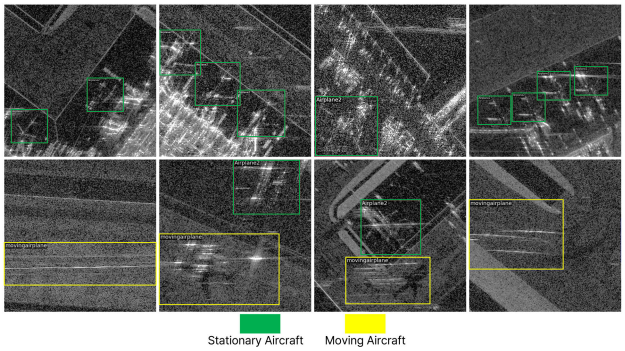


Fig. 3. Visualized examples from the aircraft dataset. The first row illustrates stationary aircraft, while the second row describes moving aircraft.

For experiment data preparation, a total of 141 SAR images, acquired between 2023 and 2024, were utilized. These data were collected from four different airports and different periods. Due to the device memory limitations, each SAR image was cropped into patches of 512×512 sizes, and the resulting dataset was randomly divided into training, validation, and test sets at a ratio of 7:2:1, respectively. Table II summarizes the number of stationary and moving aircraft instances in each subset, highlighting the class imbalance present in the dataset.

Fig. 3 presents visualized examples from the experimental dataset. The distribution of aircraft dimensions, including the

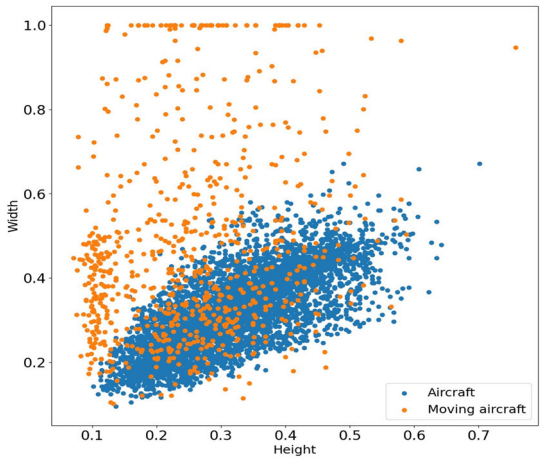


Fig. 4. Illustration of the width and height ratio distribution of stationary and moving aircraft. All the labels were processed by rigorous manual comparison from skilled experts, along with ADS-B data matching.

ratio between height and width, is analyzed in Fig. 4. Although stationary aircraft demonstrate a relatively linear width-to-height ratio, moving aircraft vary in their dimension due to the motion-derived phase defocusing effect. In cases where the aircraft is moving at very high velocity, such as during takeoff or landing, the target signatures appear highly elongated in either the width or height direction.

B. Data Augmentation Based on Target Velocity Phase Distortion

Unlike stationary targets, moving targets are subject to phase distortions, as the SAR SLC phase is modified due to the target velocity [26]. SAR slant range equation is presented as in (1), where η, R, and y_0 indicate azimuth time, slant, and ground range, respectively, while v_a, v_r and v_s signify each azimuth, range, and satellite velocity [23], [26].

Accordingly, the phase of moving target is expressed as (2). When the aircraft is moving with velocity, namely $v_a \neq 0$, the azimuth velocity component modifies the second-order term of η in $R(\eta)$ and the Doppler rate, which is a derivative of the Doppler history

$$R(\eta) \cong R + \frac{y_0 v_r}{R}\eta + \frac{1}{2R}\left((v_s - v_a)^2\right)\eta^2 \tag{1}$$

$$\phi(\eta) = -\frac{4\pi}{\lambda}R - \frac{4\pi}{\lambda}\frac{y_0 v_r}{R}\eta + \frac{2\pi}{\lambda R}(v_s - v_a)^2\eta^2. \tag{2}$$

Additionally, the first-order term of η induces an azimuth offset δ_a which shifted the target toward azimuth direction [51] as denoted in the following equation:

$$\delta_a = -\frac{v_r y_0}{v_s}. \tag{3}$$

In this case, the target position is displaced without any deformation in the target signature. Consequently, the additional phase contributions relative to the reference phase modified the target appearance depending on the direction of target motion.

The target velocity phase refocusing function proposed by [26] is able to intentionally simulate phase distortions subject to a certain azimuth velocity as denoted in (4).

To augment the training data, the moving aircraft samples are synthetically modified by utilizing the refocusing function to generate the phase-defocused aircraft images as in Fig. 5(a). The selected velocity range is grounded in the operational speed characteristics of aircraft during taxiing, takeoff, and landing phases, which constitute the primary scenarios captured in airport-area SAR imagery. According to the ICAO approach [52], speed categorization, civil airliners exhibit final approach and takeoff speeds of approximately 59–82 m/s, while larger military aircraft reach up to approximately 95 m/s during approach phases. Therefore v_a are randomly sampled within the range from −90 to 90 m/s at 1 m/s intervals to simulate varying degrees of motion-induced phase distortion

$$\varphi_{simulated}(f_\tau, f_\eta) = \frac{\pi\lambda R}{2\left(1 + \frac{\lambda f_\tau}{c0}\right)}\left(\frac{1}{v_s^2} - \frac{1}{(v_s - v_a)^2}\right)f_\eta^2. \tag{4}$$

However, defocused targets in the SAR images introduce significant distortions in the background clutter. Despite the augmentation of the SAR aircraft training data, the aircraft detection performance is able to deteriorate due to implausibly defocused clutter conditions. This is able to be remedied by considering the realistic background clutter conditions. We employ grounded-language-to-image generation (GLIGEN) [27] to suppress azimuth offset caused by range velocity effects; defocused clutter with diverse background scenes is replaced, following the approach outlined in [53]. The GLIGEN generates objects on the input image by using multiple conditions, including class, location, and prompt. The location condition l_{bbox}, converted into an embedding vector using the Fourier transform F, is grounded with an embedding text feature f_{text} by multilayer perceptron $MLP(\cdot,\cdot)$ denoted in (5). Through gated self-attention with grounded condition g^e, the input SAR imagery is transformed into synthesized data enriched with learned visual features

$$g^e = MLP\left(f_{text}, \mathcal{F}\left(l_{bbox}\right)\right). \tag{5}$$

The moving aircraft with range velocity are widely observed in the regions where aircraft are not expected to be located, beyond runways or taxiways. For their simulation, we set the text prompt to describe the situation of the aircraft as illustrated in Fig. 5(b). The auxiliary information from prompts such as "taxiing," "landing or takeoff" and "near airport" is encoded into embedding vectors that enable the generation of diverse backgrounds.

We synthesize the δ_a effect using the GLIGEN applied to the defocused SAR images containing azimuth velocity- derived-phase distortions. This diffusion-based model learns diverse background scenes from textual captions and uses them to replace the defocused background clutter, generating the GEC-like outputs that reflected velocity-induced phase effects.

The augmented data from these stages are then integrated to account for both azimuth and range velocity effects, as shown in Fig. 5(c). As a result of these sequential processes, the amount of synthesized data for moving aircraft increases by approximately 21.6%.

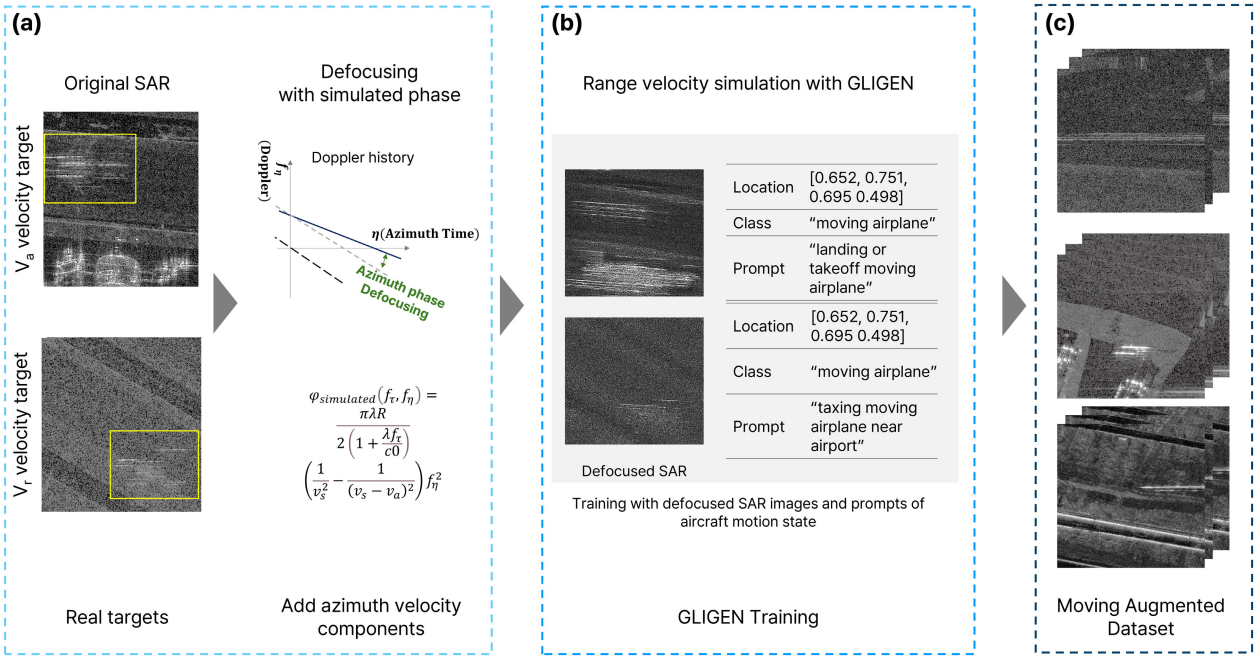


Fig. 5. Schematic illustration of the augmented data synthesis process. (a) Incorporation of azimuth velocity effects, (b) replacement of defocused background clutter using GLIGEN, and (c) resulting augmented data.

C. Framework of Aircraft Detector

This study adopts DINO [54] as a baseline model, a transformer-based detector that eliminates reliance on handcrafted components like nonmaximum suppression (NMS).

In addition, the DINO detector demonstrates accurate detection performance even in the complicated scenes, especially when targets are superposed. Leveraging the attention mechanisms, this detector effectively identifies targets in the complex background, containing artificial structures, even those that may lead to backscatter-induced false alarms. This capability ensures robust performance under different image acquisition conditions [55], [56]. Transformer-based detectors, such as DETR and DINO, directly predict both class and location from their feature maps.

Nevertheless, the detection accuracy is able to be degraded when the target features are insufficiently captured [55]. Therefore, the abundance feature map must be delivered to the detectors. In contrast to stationary aircraft, the moving aircraft present significantly different target dimensions in both width and length due to motion-induced phase distortions. As the velocity of the aircraft increases, the defocusing effect becomes intensified, leading to greater variation in the apparent signatures of moving aircraft. Consequently, the aircraft with different velocities indicated distinct phase-distorted signatures, which the backbone network often fails to effectively capture. Therefore, we propose a 2-D module that enhances feature information from the backbone shown in Fig. 6.

The structure of the 2-D module is illustrated in Fig. 7. First, this study adopts deformable convolution (DCN) [57] to tackle various shapes from the motion state target. While vanilla convolution uses fixed-size kernels to extract feature maps, DCN applies flexible kernels by adding a 2-D offset. Given a 3×3 convolution layer, the filter region $\mathcal{R}$ could be described as follows:

$$\mathrm{R} = \begin{Bmatrix} (1,1) & (1,0) & (1,-1) \\ (0,1) & (0,0) & (0,-1) \\ (-1,1) & (-1,0) & (-1,-1) \end{Bmatrix}. \tag{6}$$

When the 2-D offset Δp_n is added to the origin points $p_0 + p_n$, each location of $\mathcal{R}$, the region corresponding to feature maps, is also modified. The DCN operation is expressed as follows, where X denotes input feature map, and ω_n represents weight of each convolution layer:

$$DCN\,(p_0) = \sum_{1}^{N} \omega_n \cdot X\,(p_0 + p_n + \Delta p_n)\,. \tag{7}$$

While training proceeds, the learnable parameters ω_n and Δp_n are optimized simultaneously, resulting in a deformable and nonuniform receptive field. The resulting output feature map, $DCN(p_0)$, facilitates the extraction of motion-induced features across targets of varying widths and heights by incorporating the offset parameters. In addition, the moving targets with high velocity illustrate defocused spatial signatures, resembling improperly focused results caused by residual phase errors during signal processing. Therefore, in such cases, the targets appear as line-shaped structures toward the azimuth direction, making it difficult for conventional backbones to extract meaningful feature representations.

We accordingly adopt a central pixel difference convolution (CPDC) [58] layer, which enhances sensitivity to directional shape distortions by computing local differences using a weight matrix $m \in \mathbb{R}^{C\times H\times W}$ as defined in the following equations:

$$CPDC\,(X) = y - y_{center} \tag{8}$$

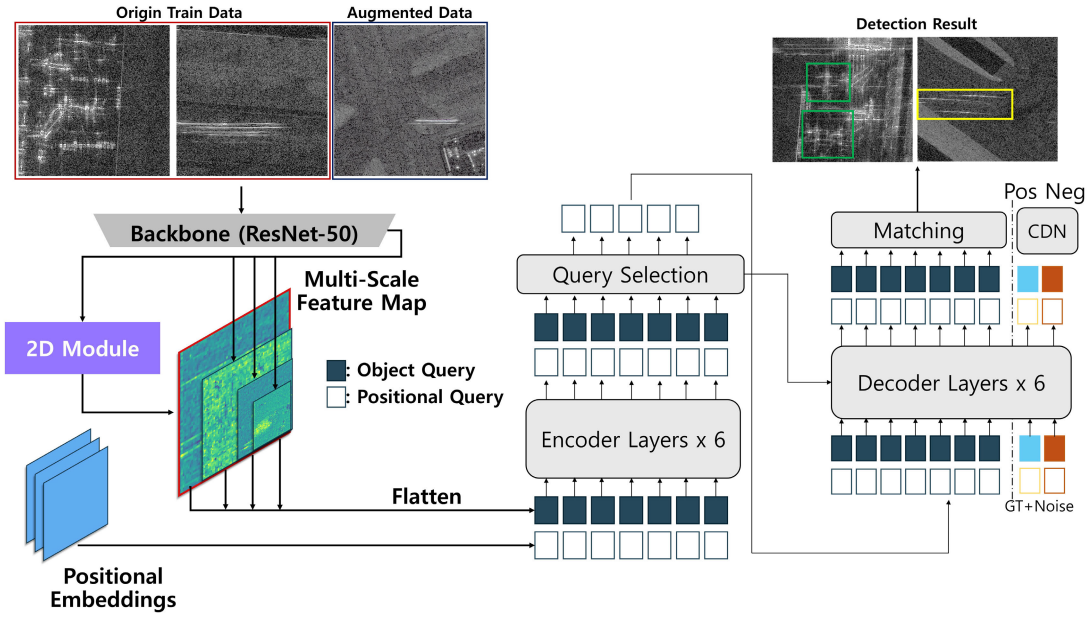


Fig. 6. Overview of the proposed aircraft detection framework.

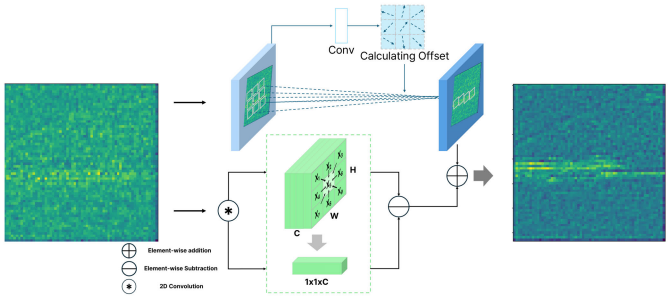


Fig. 7. Detailed structure of the 2-D module inside the aircraft detection framework.

$$y_{center} = Sum(m) * X, \quad y = m * X. \tag{9}$$

The $y_{center} \in \mathbb{R}^{C\times1\times1}$ serves a role which refines the center pixel by convolution $(*)$ between input feature map X and $Sum(m) \in \mathbb{R}^{C\times1\times1}$. By computing the difference between y and y_{center}, the differential features are extracted, and (8) is reorganized as follows:

$$CPDC(X) = f(\nabla X, \theta) = \sum_{i,j=1}^{n} \omega_{i,j} \cdot \left(X_{i,j} - X_{center}\right). \tag{10}$$

In this process, ∇x, the gradients between neighbor pixels relative to the center pixel are extracted, allowing the CPDC to train from their spatial relationships. As illustrated in Fig. 7, the edge-like signatures resulting from velocity-driven phase distortions are refined. The baseline detector utilizes multiscale features extracted by the backbone, prior to the transformer-based encoders.

Among the feature maps, the low-level feature map $F \in \mathbb{R}^{256\times64\times64}$ contains high spatial resolution and captures fundamental features of the target, such as edges and textures, whereas the high-level feature map conveys semantic information at a lower resolution. Inspired by [8], [37], and [55], both the DCN and CPDC are simultaneously applied to F, and the resulting two features are integrated via element-wise summation $(\oplus)$, thereby forming $2D(F)$, as denoted in the following equation:

$$2D(F) = CPDC(F) \oplus DCN(F). \tag{11}$$

To enhance the detector's capability to extract motion-induced features, the 2-D module is applied to the early-stage feature map.

The feature map $DCN(F)$, obtained by applying the DCN to F, retains flexible spatial representations of the targets with various dimensions. Accordingly, the feature map $CPDC(F)$, derived from the $CPDC$ layer, captures edge-like features of aircraft distorted by velocity-induced phase. By fusing these representations, the proposed framework effectively leverages phase-specific information to detect the motion state of aircraft.

IV. Experiments and Discussion

A. Detailed Experimental Conditions

1) Training Settings: Our experiments were conducted in a PyTorch-based framework; MMDETECTION and MMYOLO [59], [60]. Since a publicly available dataset for moving aircraft is not accessible, this study evaluated the proposed method using the SAR aircraft dataset constructed in Section II-A, which contains 9773 aircraft. Additionally, the ResNet-50 network [61], pretrained on ImageNet, is used as a backbone network, and a six-layer transformer encoder-decoder architecture with 256-D hidden features is employed accordingly.

The proposed framework was trained for 24 epochs based on the vanilla DINO architecture, utilizing an AdamW [62]

TABLE III
PARAMETERS CONFIGURATION OF AIRCRAFT DETECTOR TRAINING

Hyper Parameter	Value
Learning rate	0.0001
Epoch	24
Encoder/Decoder number	6
Query number	300
Feature map number	4
CIoU loss coefficient	2.0
L1 loss coefficient	5.0
Focal loss coefficient	2.0
Label noise scale	0.5
Box noise scale	1.0

optimizer with weight decay regularization and an initial learning rate of 0.0001. The learning rate was decayed by a factor of 0.1 at the 11th and 23rd epochs using a multistep learning rate schedule. The experiments were conducted using CUDA 11.2 and PyTorch 1.8.1 on Ubuntu 18.04. Two NVIDIA GeForce RTX 3090 GPUs with 24 GB of RAM were utilized. Detailed parameter settings are described in Table III.

2) Loss Function: To calculate classification and localization losses, we used three different integrated loss functions: focal loss [63], L1 loss, and complete intersection over union (CIoU) loss [64]. Each loss is calculated as follows:

$$L_{cls} = \alpha \sum_{i=1}^{n} (1 - p_i)^{\gamma} \log(p_i) \tag{12}$$

$$L_{bbox} = \frac{1}{n} \sum_{i=1}^{n} |GT_i - Pred_i| \tag{13}$$

$$L_{CIoU} = 1 - IoU + \frac{\rho^2 \left(bbox_{pred}, bbox_{GT}\right)}{c^2} + av. \tag{14}$$

In (12) and (13), where p_i denotes the classification score of the ith sample, GT_i and $Pred_i$ represent the ground truth and predicted bounding boxes, respectively. The $\rho^2(bbox_{pred}, bbox_{GT})$ refers to the squared Euclidean distance between the centers of the predicted and ground truth bounding boxes. In (14), the variable c indicates the diagonal length of the smallest enclosing box that covers both bounding boxes. Lastly, av denotes the product of a trade-off parameter a, and the aspect ratio consistency term v. In L_{cls}, the $\alpha = 0.25$ and $\gamma = 2$ were set for calculating classification loss, while each parameter controlled the impact of the imbalanced samples. L_{CIoU} considers the overlapping area, center points distance, and aspect ratio, respectively, according to (14). Therefore, the total loss of the proposed method is estimated by combining each loss term with its corresponding weight accordingly, as follows:

$$L_{total} = 2.0 * L_{cls} + 5.0 * L_{bbox} + 2.0 * L_{CIoU}. \tag{15}$$

3) Evaluation Metrics: For quantitative evaluation, the generally used metric average precision (AP) is employed. $mAP_{0.5}$ and $mAP_{0.5:0.95}$ are utilized according to (16), which corresponds to the area of the precision (P) and recall (R) curve. The mAP is defined as follows:

$$mAP = \int_0^1 P(R) dR, \quad \text{IoU} = 0.5 \text{ and } 0.5 \text{ to } 0.95 \tag{16}$$

TABLE IV
ABLATION STUDY OUTCOME OF THE PROPOSED AIRCRAFT DETECTOR

Setting \ Eval	mAP $_{0.5:0.95}$	Aircraft $AP_{0.5}$	Moving aircraft $AP_{0.5}$	Moving aircraft $AP_{0.5:0.95}$
DINO	0.459	0.932	0.800	0.396
+DCN	0.459	0.932	0.771	0.395
+CPDC	0.468	0.931	0.795	0.407
+2D Module	0.463	0.923	0.831	0.407
+ augmented data (without GLIGEN)	0.476	0.93	0.836	0.421
+ augmented data (with GLIGEN)	0.482	0.933	0.836	0.442
+augmented Data + 2D Module	**0.492**	**0.934**	**0.855**	**0.462**

where P and R denote precision and recall, and are calculated as (17) and (18), respectively; TP, FP, and FN indicate true positive, false positive, and false negative

$$\text{Precision} = \frac{TP}{TP + FP} \tag{17}$$

$$\text{Recall} = \frac{TP}{TP + FN}. \tag{18}$$

B. Ablation Study

This study conducted comprehensive ablation experiments. Based on the baseline model, DINO, the 2-D module and the augmented data were incorporated separately to verify the effect on model performance advancement. The parameter settings were fixed during the test for variable control. The effect of adding or removing each element on the evaluation metrics is presented in Table IV.

1) Augmentation Strategy: To validate the effectiveness of the proposed augmentation strategy, we compared the baseline evaluation metrics both with and without the augmented data, encompassing both cases with and without the utilization of the GLIGEN. As shown in Table III, adopting augmented data enhances the moving aircraft $AP_{0.5}$ from 0.80 to 0.836 and $AP_{0.5:0.95}$ from 0.396 to 0.462. In the case of not applying GLIGEN, the $AP_{0.5}$ and $AP_{0.5:0.95}$ are enhanced with 3.6% and 2.5%. In addition, when adopting the GLIGEN, $AP_{0.5:0.95}$ is improved with 2.1% advancement compared to the case without GLIGEN. While the detector trained without GLIGEN-augmented data struggles to accurately localize moving targets due to background distortions, the model trained with GLIGEN effectively mitigates such artifacts, resulting in improved performance in $AP_{0.5:0.95}$ of moving aircraft. This indicates that the use of GLIGEN for data augmentation not only suppresses defocusing of clutter but also successfully simulates azimuthal displacement, thereby accurately replicating the physical phenomena observed in real moving aircraft.

The qualitative detection results further demonstrate the importance of the augmented data, which were synthesized

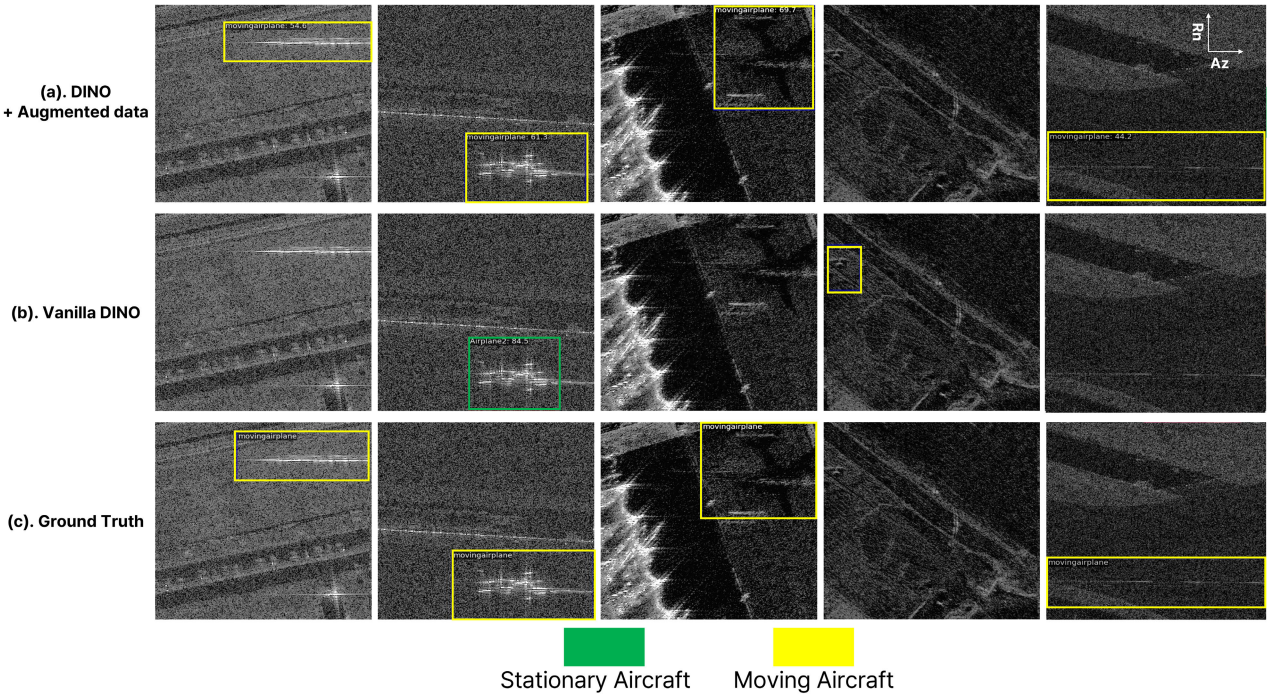


Fig. 8. Detection results with and without the use of augmented data. Green and yellow boxes represent stationary and moving aircraft, respectively. Each row indicates (a) results from DINO trained with the proposed augmentation strategy, (b) results from vanilla DINO, and (c) ground-truth annotations.

using simulated azimuth-velocity phase and the GLIGEN-based azimuth offset effects. As illustrated in Fig. 8(a) and (b), the baseline model exhibits poor detection performance for the aircraft in motion regardless of background clutter. This is caused by the detector not being trained for motion-induced characteristics due to data scarcity. As shown in Fig. 8(b), the augmented data generated using the proposed strategy significantly improves detection performance by enabling the detector to be trained from a wider range of motion states. Furthermore, false positives in background regions that resemble moving aircraft; however, these errors are largely eliminated in the model trained using the augmented dataset. This result indicates that the proposed target augmentation strategy is effective in resolving the data scarcity problem.

2) 2-D Module: The effect of the 2-D module for refining the characteristic features from the moving aircraft is accordingly investigated. We first analyzed the individual contributions of DCN and CPDC, as the proposed 2-D module is composed of both layers. When using only the DCN, the $AP_{0.5:0.95}$ for moving aircraft remained unchanged, but the $AP_{0.5}$ decreased compared to the baseline. In contrast, applying CPDC alone resulted in a 1.1% improvement in $AP_{0.5:0.95}$, although the $AP_{0.5}$ slightly dropped to 0.795 (a 0.5% decrease). However, when utilizing the 2-D module, the $AP_{0.5}$ improves significantly by 3.6% as represented in Table IV. These results suggest that the various-scale features captured by the DCN contribute synergistically only when fused with the edge-enhanced representations extracted by the CPDC. Building on this synergy, the proposed network is able to extract feature maps that are otherwise difficult to capture with conventional backbone architectures, particularly for moving aircraft with diverse spatial extents.

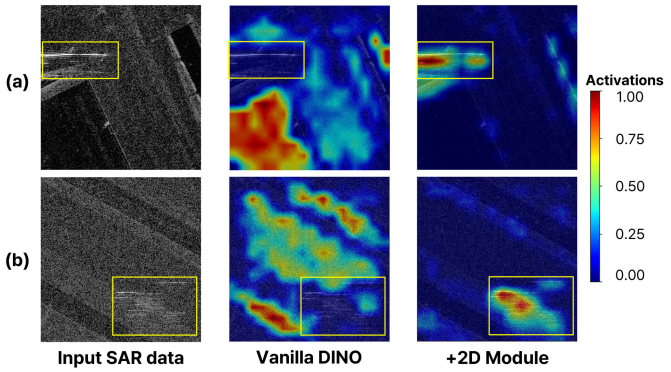


Fig. 9. Eigen-CAM visualizations of DINO and DINO with 2-D module. (a) Activations for a target with azimuth-direction velocity. (b) Activations for a target with range-direction velocity. Red regions indicate higher activation levels.

TABLE V
IMPACT OF EACH FEATURE LEVEL ON DETECTION ACCURACY

Stage \ Eval	mAP $_{0.5:0.95}$	Aircraft $AP_{0.5}$	Moving aircraft $AP_{0.5}$	Moving aircraft $AP_{0.5:0.95}$
1	**0.492**	0.934	**0.855**	**0.462**
2	0.470	0.933	0.822	0.425
3	0.484	**0.935**	0.851	0.442
4	0.482	0.929	0.845	0.437

To further illustrate the effectiveness of the 2-D module, a supplementary analysis is conducted with Eigen CAM [65] as shown in Fig. 9. The target with significant azimuth motion is presented in Fig. 9(a), where the vanilla DINO fails to

TABLE VI
COMPARISON OF DETECTION PERFORMANCE ON SELF-BUILT DATASET. THE BEST RESULT IS MENTIONED BOLD

Model \ Eval	mAP		Aircraft		Moving aircraft		Parameters (M)	FLOPs (G)	FPS
	0.5	0.5:0.95	0.5	0.5:0.95	0.5	0.5:0.95			
Cascade R-CNN	0.856	0.461	0.909	0.525	0.803	0.398	69.395	88.60	19.0
Sparse R-CNN	0.832	0.429	0.910	0.496	0.755	0.362	77.800	**23.28**	23.0
YOLOv8X	0.719	0.388	0.863	0.492	0.575	0.283	68.200	257.80	**57.1**
RepPoints	0.837	0.442	0.906	0.506	0.768	0.379	36.822	48.43	20.0
DQ-DETR	0.831	0.466	0.919	0.525	0.740	0.410	59.000	902.50	4.7
YOLOv12L	0.795	0.450	0.905	**0.542**	0.684	0.359	**26.340**	88.60	40.8
DINO	0.860	0.459	0.932	0.523	0.800	0.396	47.391	167.48	18.9
Ours	**0.895**	**0.492**	**0.934**	0.523	**0.855**	**0.462**	48.632	168.04	18.4

localize moving aircraft. In contrast, the proposed framework shows highly activated localization performance despite the defocused shapes. Fig. 9(b) represents the aircraft subject to an azimuth offset caused by range velocity. This module accurately captures feature information, while the baseline feature map fails to focus on the targets. This result highlights the edge information extraction competence of the 2-D module while maintaining the detection accuracy of stationary targets.

3) Impact of Each Feature Level on Detection Accuracy: In the proposed aircraft detector, the 2-D module is applied to the low-level feature maps to refine velocity-induced characteristics. We analyzed which level of the feature map has the most notable impact on detection accuracy. Table V presents a comparison of evaluation metrics across different feature map levels. Regardless of the stage, the 2-D module improves detection performance compared to the baseline detector. The experiment results demonstrate the highest mAP, $AP_{0.5,}$ and $AP_{0.5:0.95}$ from moving aircraft, respectively, when applying the 2-D module to the lowest level feature map. The evaluation metrics are presented in the order corresponding to the feature maps from the third, fourth, and second stages, respectively. This indicates that the phase information from the target velocity is precisely captured.

Hence, incorporating the 2-D module and augmentation strategy results in a moving aircraft $AP_{0.5}$ of 0.855, representing a 5.5% improvement, and an $AP_{0.5:0.95}$ of 0.462, reflecting a 6.6% improvement. Each component of the proposed framework contributes to detection performance enhancement, particularly for the moving aircraft, while maintaining high accuracy for the stationary targets. This improvement further demonstrates that the 2-D module effectively refines feature maps of defocused targets, and that the augmented data enables the detector to learn motion-induced phase distortion characters by increasing the diversity and quantity of training data.

C. Detection Results

To evaluate the proposed aircraft detector in detail, we compared several deep-learning object detection algorithms, including two-stage and one-stage detectors; Cascade R-CNN [66], Sparse R-CNN [67], YOLOv8 [68], RepPoints [69], DQ-DETR [70], and YOLOv12 [71].

According to Table VI, the detection results show that our proposed framework achieves superior performance in the $AP_{0.5}$ and $AP_{0.5:0.95}$, especially in detecting the moving aircraft. Compared with conventional two-stage models: Cascade R-CNN and Sparse R-CNN, the proposed method improves 3.9%, 6.3% in $mAP_{0.5,}$ and 3.1%, 6.3% in $mAP_{0.5:0.95,}$ respectively. Similarly, in comparison to the single-stage detectors: YOLOv8 and RepPoints, our method demonstrates significant improvement of 17.6%, 5.8% in $mAP_{0.5}$. Compared to attention-based detectors such as DQ-DETR and YOLOv12, the proposed method demonstrates competitive performance, achieving improvements of 5.2% and 10.3%, respectively, in the $AP_{0.5:0.95}$ for moving aircraft.

While two-stage detectors outperform one-stage detectors, the CNN-based algorithms fail to attain optimal accuracy in the experiments, primarily due to the lack of feature extraction competence of detectors. Even in attention mechanisms, the insufficient enhancement of low-level features related to moving targets extracted by the backbone results in suboptimal performance for detecting moving aircraft. In contrast, both the DINO and the proposed framework demonstrate that transformer-based architectures are able to deliver robust performance, even in the presence of challenging background clutter.

When comparing the baseline DINO with the proposed detector, performance enhancement is evident for recognition of moving aircraft with an advancement of 5.5% in $mAP_{0.5}$ and 6.6% in $mAP_{0.5:0.95,}$ respectively. Fig. 10 illustrates detection results from different detectors across various background scenes, corresponding to (a)–(d), respectively. In Fig. 10(a), the two-stage detectors and YOLOv12 represent false alarms to similar patterns with defocused targets, while the DINO-based detectors represent no false detections. Fig. 10(b) is the case of moving aircraft with range velocity components on the runway. In this case, the proposed detector illustrates accurate localization despite the azimuth offset. In addition, as shown in Fig. 10(c) and (d), when the scene background contains complex background environments, conventional detectors exhibit a higher susceptibility to missed detections and false positives. Even state-of-the-art models such as DQ-DETR and YOLOv12 represent false classifications induced by background clutter. In contrast, the proposed detector, which integrates the 2-D enhance-

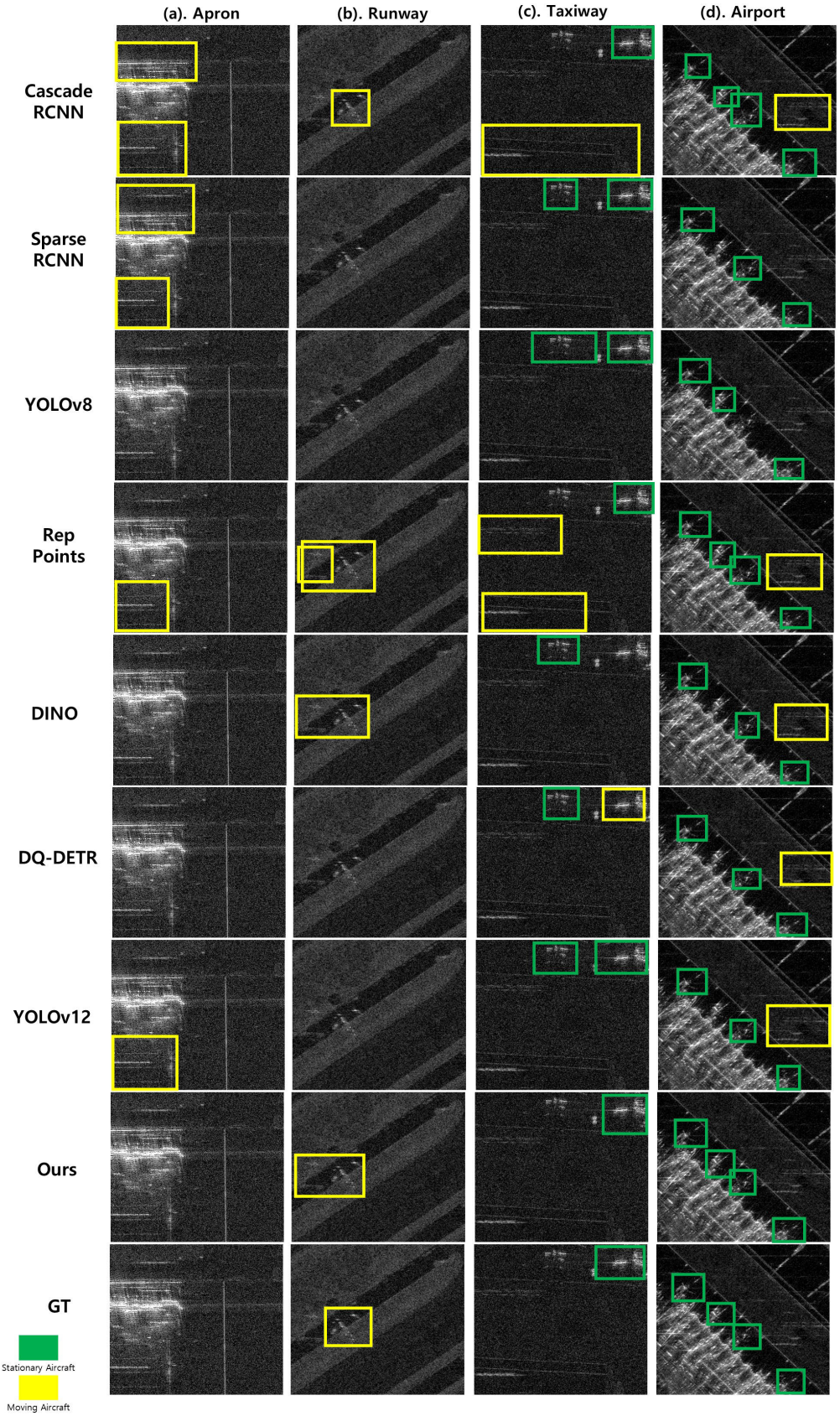


Fig. 10. Qualitative detection results of various methods in different backgrounds on the self-built dataset. (a)–(d) Illustrates diverse airport background clutter scenes.

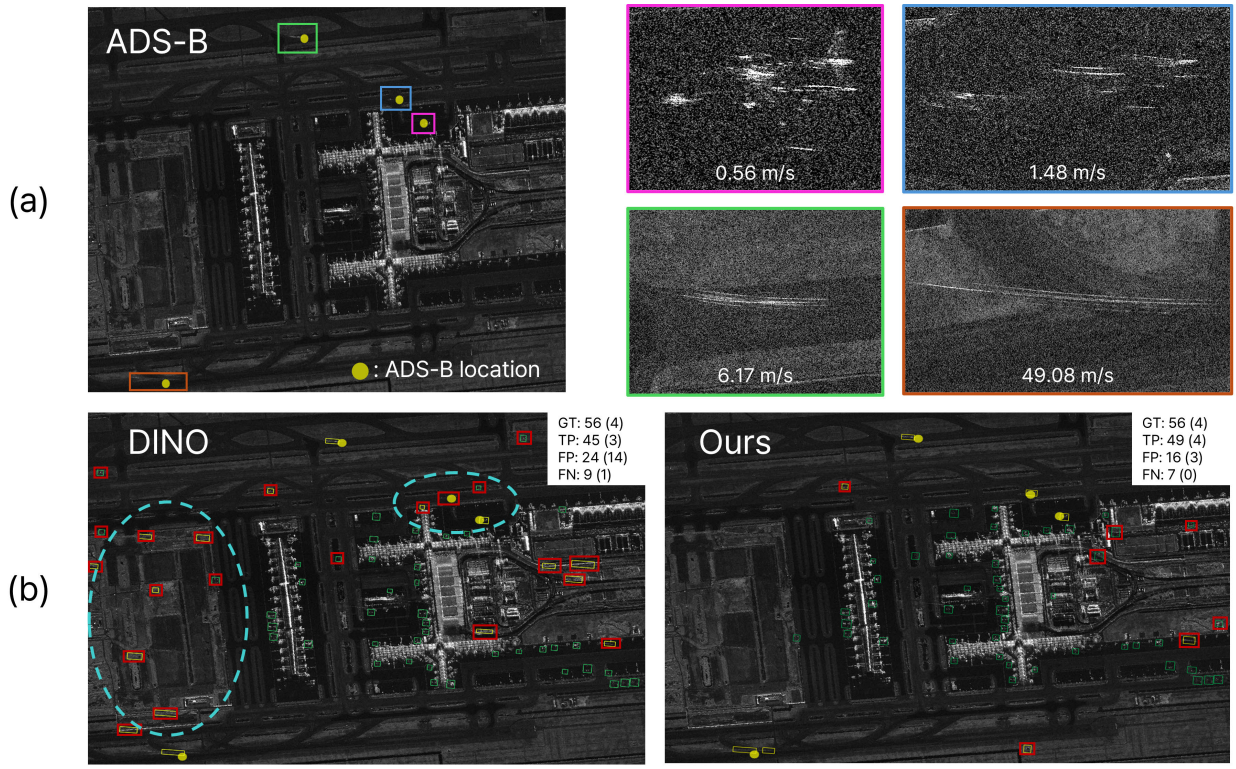


Fig. 11. Qualitative detection results on a real large SAR scene. (a) SAR image with overlaid ADS-B (yellow circles) and the shape of each aircraft in motion. (b) Detection results from the baseline DINO and the proposed framework. The number in parentheses indicates the total number of moving aircraft in the scene. Dotted circles indicate incorrect detections from vanilla DINO compared to the proposed method. Yellow boxes indicate moving aircraft, green boxes denote stationary aircraft, and red boxes represent false alarms or missed detections.

ment module and the augmented training dataset, consistently outperforms under such challenging conditions. In summary, unlike the other detectors, including the vanilla DINO, our detector is robust under different airport background scenes, demonstrating that the modifications in the detection framework significantly enhance the baseline's performance.

D. Case Study

Furthermore, we investigated the practical applicability of the proposed aircraft detector, especially whether it demonstrates high performance under various target velocity conditions. As mentioned above, the ADS-B data are used to validate the motion state of the targets. Fig. 11 illustrates the detection results at Suvarnabhumi Airport, Thailand, acquired on January 13, 2025, with ADS-B data. Fig. 11(a) denotes the shapes of moving aircraft under diverse situations: taxiing and landing, and a large scene overlayed with interpolated ADS-B. The interpolated ADS-B signals are accurately matched with the moving targets, while the stationary aircraft are validated through visual inspection. In addition, the appearance of aircraft in motion varies significantly; as velocity increases, the phase defocusing effect becomes severe. For instance, in landing scenarios, the aircraft appears as a linear streak with 49.08 m/s velocity.

Fig. 11(b) indicates the detection results on the real full-scene SAR from the DINO and the proposed algorithm. The yellow and green boxes denote the stationary aircraft and motion state target, respectively, while the red boxes indicate false alarms and missed detections. It is clear that our proposed framework has fewer false alarms, while vanilla DINO represents 38 false detections in backscatter patterns similar to moving aircraft. Additionally, although the baseline fails to detect taxiing aircraft moving at 1.48 m/s, the proposed algorithm demonstrates robust performance in moving aircraft detection regardless of the velocity conditions. When the aircraft is in motion, the signature of the aircraft exhibits complex backscattering characteristics, which complicate the reliable discrimination of moving targets. Nevertheless, the proposed framework effectively leverages motion-induced phase features by integrating the augmented training data and the 2-D module, thereby enabling robust target detection under real-world operational conditions. This indicates that the proposed 2-D module and the data augmentation strategy based on the synthesized targets are effective in enhancing the overall information for moving aircraft, while maintaining the detection accuracy for stationary ones.

E. Generalizability and Future Work

The proposed moving aircraft detection method focuses on the phase distortion information induced by target velocity components. The azimuth defocusing and azimuth displacement of moving aircraft produce signatures that differ substantially from those of stationary aircraft in both shape and contextual information, causing detectors to suffer from false alarms and missed detections. The proposed framework addresses these challenges by enhancing motion-induced

shape features through the 2-D module and overcoming contextual distortions arising from positional errors via a phase distortion-based data augmentation strategy. Importantly, the two core components of the proposed framework do not rely on specific wavelengths, polarimetric information, or particular acquisition modes such as Spotlight mode; instead, they are grounded in Doppler phase information, which is fundamental to the SAR image formation process. Therefore, the proposed framework is expected to operate robustly not only with the Umbra satellite used in this study, but also across other platforms with different sensor parameters, such as ICEYE, Gaofen-3, and ALOS. In contrast, applying the proposed method to medium-resolution SAR datasets such as Sentinel-1 and AIRSAR, which provide a spatial resolution of approximately 10 m, remains challenging because aircraft are represented by only a few pixels and motion-induced defocusing artifacts are less distinctly manifested at coarser resolutions, thereby reducing the discriminability of moving aircraft signatures that the proposed method relies upon. Extending the framework to such medium-resolution datasets is therefore left for future investigation. It should be noted, however, that since the augmentation process requires phase information to synthesize phase-distorted moving targets, SLC products that provide Doppler phase information are required, in addition to backscattering amplitude products such as GEC images.

Furthermore, while this study focuses on moving aircraft detection, the proposed method is principally applicable to other target types, as it is based on Doppler phase information common to all moving targets. In particular, the framework is expected to perform effectively for targets such as vessels, where shape distortion and azimuth offset have a significant influence on detection. The method is also applicable to vehicles exhibiting similar phase distortion phenomena; however, given their relatively small physical dimensions, the degradation of signal-to-noise ratio (SNR) during azimuth defocusing must be jointly considered, and validation for such targets will be addressed in future work.

V. Conclusion

This article proposes an accurate aircraft detector for high-resolution SAR, presenting a robust detection performance on moving targets. To remedy the velocity-induced phase distortion and complex background interference, the 2-D module and data augmentation strategy are introduced, incorporating the transformer detector DINO. In the ablation study, we verified the importance of the 2-D module and augmenting the training data generated with the GLIGEN and target motion phase refocusing function. Substantial comparative experiments using various CNN architectures on the self-constructed dataset demonstrated the superiority and practicality of the proposed detector in detecting and recognizing aircraft targets. Moreover, the real application into full-scene SAR demonstrated that our proposed detector was effectively applied to detecting and recognizing aircraft targets regardless of different velocities, which effectively suppresses false alarm detections.

Subsequent studies will focus on identifying the aircraft motion situations, such as taxiing, landing and taking-off situations.

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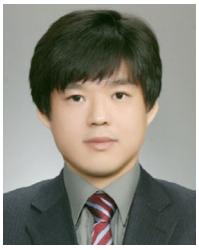
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